

RLS VS VIEW

Query No 4:

```
SELECT
    o_orderpriority,
    COUNT(*) AS order_count
FROM orders
WHERE o_orderdate >= DATE '1994-01-01'
    AND o_orderdate < DATE '1994-01-01' + INTERVAL '3' MONTH
    AND EXISTS (
        SELECT *
        FROM lineitem
        WHERE l_orderkey = o_orderkey
            AND l_commitdate < l_receiptdate
    )
GROUP BY o_orderpriority
ORDER BY o_orderpriority;
```

Policy Applied:

```
CREATE POLICY orders_policy
ON orders
FOR SELECT
USING (
    EXISTS (
        SELECT 1
        FROM customer c
        WHERE c.c_custkey = orders.o_custkey
            AND c.c_acctbal > 5000
    )
);
```

Planning Time: 0.512 ms
Execution Time: 2576.247 ms

Flow to track for RLS:

```
Q4 SQL
↓
1. Raw Parse Tree
↓
2. Analyzed Query Tree
↓
3. Rewritten Query Tree
   (RLS injected / View expanded)
↓
4. Planner Input
↓
5. Final Execution Plan
```

1. Raw Parse Tree: src/backend/tcop/postgres.c

Function: pg_analyze_and_rewrite_fixedparams()

After: parsetree_list=pg_parse_query(query_string);

Print:

```
eelog(WARNING, "==== RAW PARSE TREE =====");

foreach(lc, parsetree_list)
{
    elog(WARNING, "%s",
        nodeToString(lfirst(lc)));
}
```

What to compare

RLS vs View: Should be IDENTICAL

2. Analyzed Query Tree

File: src/backend/parser/analyze.c

Function: parse_analyze_fixedparams()

After: query=transformTopLevelStmt(...);

Print:

```
eLog(WARNING,
"===== ANALYZED QUERY =====");

eLog(WARNING,
"%s",
nodeToString(query));
```

What to compare

Check:

```
TargetList
RangeTblEntry
SubLink(EXISTS)
GroupClause
```

Again RLS and View should be nearly identical.

3. Rewrite Phase (MOST IMPORTANT). This is where they diverge.

File: src/backend/rewrite/rewriteHandler.c

Function: QueryRewrite()

Before rewrite:

```
eLog(WARNING,
"===== REWRITE INPUT =====");

eLog(WARNING,
"%s",
nodeToString(parsetree));
```

After rewrite:

```
eelog(WARNING,
"===== REWRITE OUTPUT =====");

eelog(WARNING,
"%S",
nodeToString(result));
```

Extra RLS Hook

File: src/backend/rewrite/rowsecurity.c

Function: get_row_security_policies()

Print:

```
eelog(WARNING,
"===== RLS POLICY ADDED =====");

eelog(WARNING,
"Table OID=%u",
rte->relid);

eelog(WARNING,
"%S",
nodeToString(*securityQuals));
```

Compare

RLS: orders + EXISTS(customer)

View: No RLS injection

4. Planner Input

File src/backend/optimizer/plan/planner.c

Function: standard_planner()

Beginning:

```
eelog(WARNING,
"===== PLANNER INPUT =====");

eelog(WARNING,
"%S",
nodeToString(parse));
```

Compare

This is the most important comparison.

Create:

```
RLS Planner Input
View Planner Input
```

5. Final Plan

File: src/backend/optimizer/plan/planner.c

Near return of: standard_planner()

Print:

```
eelog(WARNING,
"===== FINAL PLAN =====");

eelog(WARNING,
"%S",
nodeToString(result->planTree));
```

```
tpch=> EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT
    o_orderpriority,
    COUNT(*) AS order_count
FROM orders
```

```

WHERE o_orderdate >= DATE '1994-01-01'
      AND o_orderdate < DATE '1994-01-01' + INTERVAL '3' MONTH
      AND EXISTS (
          SELECT 1
          FROM lineitem
          WHERE l_orderkey = o_orderkey
                AND l_commitdate < l_receiptdate
      )
GROUP BY o_orderpriority
ORDER BY o_orderpriority;
WARNING:  ===== RAW PARSE TREE =====
WARNING:  {RAWSTMT :stmt {EXPLAINSTMT :query {SELECTSTMT :dis
tinctClause <> :intoClause <> :targetList ({RESTARTGET :name <
> :indirection <> :val {COLUMNREF :fields ("o_orderpriority")
:location -1} :location -1} {RESTARTGET :name order_count :ind
irection <> :val {FUNCCALL :funcname ("count") :args <> :agg_
order <> :agg_filter <> :over <> :agg_within_group false :agg
_star true :agg_distinct false :func_variadic false :funcform
at 0 :location -1} :location -1}) :fromClause ({RANGEVAR :cat
alogname <> :schemaname <> :relname orders :inh true :relpers
istence p :alias <> :location -1}) :whereClause {BOOLEXPR :bo
olop and :args ({A_EXPR :name (">=") :lexpr {COLUMNREF :field
s ("o_orderdate") :location -1} :rexpr {TYPECAST :arg {A_CONST
T :val "\1994-01-01" :location -1} :typeName {TYPENAME :names
("date") :typeOid 0 :setof false :pct_type false :typmods <>
:typemod -1 :arrayBounds <> :location -1} :location -1} :rexp
r_list_start -1 :rexpr_list_end -1 :location -1} {A_EXPR :nam
e ("\<") :lexpr {COLUMNREF :fields ("o_orderdate") :location
-1} :rexpr {A_EXPR :name ("+") :lexpr {TYPECAST :arg {A_CONST
:val "\1994-01-01" :location -1} :typeName {TYPENAME :names
("date") :typeOid 0 :setof false :pct_type false :typmods <>
:typemod -1 :arrayBounds <> :location -1} :location -1} :rexp
r {TYPECAST :arg {A_CONST :val "\3" :location -1} :typeName
{TYPENAME :names ("pg_catalog" "interval") :typeOid 0 :setof
false :pct_type false :typmods ({A_CONST :val 2 :location -
1}) :typemod -1 :arrayBounds <> :location -1} :location -1} :

```

```

rexpr_list_start -1 :rexpr_list_end -1 :location -1} :rexpr_list_start -1 :rexpr_list_end -1 :location -1} {SUBLINK :subLinkType 0 :subLinkId 0 :testexpr <> :operName <> :subselect {SELECTSTMT :distinctClause <> :intoClause <> :targetList ({RETARGET :name <> :indirection <> :val {A_CONST :val 1 :location -1} :location -1}) :fromClause ({RANGEVAR :catalogname <> :schemaname <> :relname lineitem :inh true :relpersistence p :alias <> :location -1}) :whereClause {BOOLEXPRESS :boolop and :args ({A_EXPR :name ("=") :lexpr {COLUMNREF :fields ("l_orderkey") :location -1} :rexpr {COLUMNREF :fields ("o_orderkey") :location -1} :rexpr_list_start -1 :rexpr_list_end -1 :location -1} {A_EXPR :name ("<") :lexpr {COLUMNREF :fields ("l_commitdate") :location -1} :rexpr {COLUMNREF :fields ("l_receiptdate") :location -1} :rexpr_list_start -1 :rexpr_list_end -1 :location -1}) :location -1} :groupClause <> :groupDistinct false :havingClause <> :windowClause <> :valuesLists <> :sortClause <> :limitOffset <> :limitCount <> :limitOption 0 :lockingClause <> :withClause <> :op 0 :all false :larg <> :rarg <>} :location -1}) :location -1} :groupClause ({COLUMNREF :fields ("o_orderpriority") :location -1}) :groupDistinct false :havingClause <> :windowClause <> :valuesLists <> :sortClause ({SORTBY :node {COLUMNREF :fields ("o_orderpriority") :location -1} :sortby_dir 0 :sortby_nulls 0 :useOp <> :location -1}) :limitOffset <> :limitCount <> :limitOption 0 :lockingClause <> :withClause <> :op 0 :all false :larg <> :rarg <>} :options ({DEFELEM :defnamespace <> :defname analyze :arg <> :defaction 0 :location -1} {DEFELEM :defnamespace <> :defname verbose :arg <> :defaction 0 :location -1} {DEFELEM :defnamespace <> :defname buffers :arg <> :defaction 0 :location -1})) :stmt_location -1 :stmt_len -1}

```

WARNING: ===== ANALYZED QUERY =====

```

WARNING: {QUERY :commandType 6 :querySource 0 :canSetTag true :utilityStmt {EXPLAINSTMT :query {QUERY :commandType 1 :querySource 0 :canSetTag true :utilityStmt <> :resultRelation 0 :hasAggs true :hasWindowFuncs false :hasTargetSRFs false :hasSubLinks true :hasDistinctOn false :hasRecursive false :hasMo

```

```

difyingCTE false :hasForUpdate false :hasRowSecurity false :h
asGroupRTE true :isReturn false :cteList <> :rtable ({RANGETB
LENTY :alias <> :eref {ALIAS :aliasname orders :colnames ("o
_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orde
rdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_commen
t"}} :rtekind 0 :relid 16561 :inh true :relkind r :rellockmod
e 1 :perminfoindex 1 :tablesample <> :lateral false :inFromCl
true :securityQuals <>} {RANGETBLENTY :alias <> :eref {ALIAS
:aliasname *GROUP* :colnames ("o_orderpriority")}} :rtekind 9
:groupexprs ({VAR :varno 1 :varattno 6 :vartype 25 :vartypmod
-1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varretu
rningtype 0 :varnosyn 1 :varattnosyn 6 :location -1}) :latera
l false :inFromCl false :securityQuals <>}) :rteperminfos ({R
TEPERMISSIONINFO :relid 16561 :inh true :requiredPerms 2 :che
ckAsUser 0 :selectedCols (b 8 12 13) :insertedCols (b) :updat
edCols (b)}) :jointree {FROMEXPR :fromlist ({RANGETBLREF :rti
ndex 1}) :quals {BOOLEXPRESS :boolop and :args ({OPEXPRESS :opno 10
98 :opfuncid 1090 :opresulttype 16 :opretset false :opcollid
0 :inputcollid 0 :args ({VAR :varno 1 :varattno 5 :vartype 10
82 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsu
p 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location
-1} {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :co
nstlen 4 :constbyval true :constisnull false :location -1 :co
nstvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]}) :location -1} {OPEX
PRESS :opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false
:opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 5 :
vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :
varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5
:location -1} {OPEXPRESS :opno 1076 :opfuncid 2071 :opresulttype
1114 :opretset false :opcollid 0 :inputcollid 0 :args ({CONST
:consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :c
onstbyval true :constisnull false :location -1 :constvalue 4
[ 113 -9 -1 -1 -1 -1 -1 -1 ]} {CONST :consttype 1186 :constty
pmod 196607 :constcollid 0 :constlen 16 :constbyval false :co
nstisnull false :location -1 :constvalue 16 [ 0 0 0 0 0 0 0 0
0 0 0 0 3 0 0 0 ]}) :location -1}) :location -1} {SUBLINK :su

```

```

bLinkType 0 :subLinkId 0 :testexpr <> :operName <> :subselect
{QUERY :commandType 1 :querySource 0 :canSetTag true :utility
Stmt <> :resultRelation 0 :hasAggs false :hasWindowFuncs fals
e :hasTargetSRFs false :hasSubLinks false :hasDistinctOn fals
e :hasRecursive false :hasModifyingCTE false :hasForUpdate fa
lse :hasRowSecurity false :hasGroupRTE false :isReturn false
:cteList <> :rtable ({RANGETBENTRY :alias <> :eref {ALIAS :a
liasname lineitem :colnames ("l_orderkey" "l_partkey" "l_supp
key" "l_linenummer" "l_quantity" "l_extendedprice" "l_discoun
t" "l_tax" "l_returnflag" "l_linestatus" "l_shipdate" "l_comm
itdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comm
ent")}} :rtekind 0 :relid 16566 :inh true :relkind r :rellockm
ode 1 :perminfoindex 1 :tablesample <> :lateral false :inFrom
Cl true :securityQuals <>)) :rteperminfos ({RTEPERMISSIONINFO
:relid 16566 :inh true :requiredPerms 2 :checkAsUser 0 :selec
tedCols (b 8 19 20) :insertedCols (b) :updatedCols (b)}) :joi
ntree {FROMEXPR :fromlist ({RANGETBLREF :rtindex 1}) :quals
{BOOLEXPRESS :boolop and :args ({OPEXPRESS :opno 96 :opfuncid 65 :o
presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :ar
gs ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varc
ollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype
0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :var
attno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrel
s (b) :varlevelsup 1 :varreturningtype 0 :varnosyn 1 :varattn
osyn 1 :location -1}) :location -1} {OPEXPRESS :opno 1095 :opfun
cid 1087 :opresulttype 16 :opretset false :opcollid 0 :inputc
ollid 0 :args ({VAR :varno 1 :varattno 12 :vartype 1082 :vart
ypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :var
returningtype 0 :varnosyn 1 :varattnosyn 12 :location -1} {VA
R :varno 1 :varattno 13 :vartype 1082 :vartypmod -1 :varcollid
0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :v
arnosyn 1 :varattnosyn 13 :location -1}) :location -1}) :loca
tion -1}} :mergeActionList <> :mergeTargetRelation 0 :mergeJo
inCondition <> :targetList ({TARGETENTRY :expr {CONST :constt
ype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval
true :constisnull false :location -1 :constvalue 4 [ 1 0 0 0

```

```

0 0 0 0 ]} :resno 1 :resname ?column? :ressortgroupref 0 :res
origtbl 0 :resorigcol 0 :resjunk false}) :override 0 :onConfl
ict <> :returningOldAlias <> :returningNewAlias <> :returning
List <> :groupClause <> :groupDistinct false :groupingSets <>
:havingQual <> :windowClause <> :distinctClause <> :sortClau
se <> :limitOffset <> :limitCount <> :limitOption 0 :rowMarks
<> :setOperations <> :constraintDeps <> :withCheckOptions <>
:stmt_location -1 :stmt_len -1} :location -1}) :location -1}}
:mergeActionList <> :mergeTargetRelation 0 :mergeJoinConditio
n <> :targetList ({TARGETENTRY :expr {VAR :varno 2 :varattno
1 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnos
yn 1 :location -1} :resno 1 :resname o_orderpriority :ressort
groupref 1 :resorigtbl 16561 :resorigcol 6 :resjunk false} {T
ARGETENTRY :expr {AGGREF :aggfnoid 2803 :aggtype 20 :aggcolli
d 0 :inputcollid 0 :aggtranstype 0 :aggargtypes <> :aggdirect
args <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :
aggstar true :aggvariadic false :aggkind n :aggpresorted fals
e :agglevelsup 0 :aggsplit 0 :aggno -1 :aggtransno -1 :locati
on -1} :resno 2 :resname order_count :ressortgroupref 0 :reso
rigtbl 0 :resorigcol 0 :resjunk false}) :override 0 :onConfl
ict <> :returningOldAlias <> :returningNewAlias <> :returningL
ist <> :groupClause ({SORTGROUPCLAUSE :tleSortGroupRef 1 :eqo
p 98 :sortop 664 :reverse_sort false :nulls_first false :hash
able true}) :groupDistinct false :groupingSets <> :havingQual
<> :windowClause <> :distinctClause <> :sortClause ({SORTGROU
PCLAUSE :tleSortGroupRef 1 :eqop 98 :sortop 664 :reverse_sort
false :nulls_first false :hashable true}) :limitOffset <> :li
mitCount <> :limitOption 0 :rowMarks <> :setOperations <> :co
nstraintDeps <> :withCheckOptions <> :stmt_location -1 :stmt_
len -1} :options ({DEFELEM :defnamespace <> :defname analyze
:arg <> :defaction 0 :location -1} {DEFELEM :defnamespace <>
:defname verbose :arg <> :defaction 0 :location -1} {DEFELEM
:defnamespace <> :defname buffers :arg <> :defaction 0 :locat
ion -1}}) :resultRelation 0 :hasAggs false :hasWindowFuncs fa
lse :hasTargetSRFs false :hasSubLinks false :hasDistinctOn fa

```

```

lse :hasRecursive false :hasModifyingCTE false :hasForUpdate
false :hasRowSecurity false :hasGroupRTE false :isReturn fals
e :cteList <> :rtable <> :rteperminfos <> :jointree <> :merge
ActionList <> :mergeTargetRelation 0 :mergeJoinCondition <> :
targetList <> :override 0 :onConflict <> :returningOldAlias <
> :returningNewAlias <> :returningList <> :groupClause <> :gr
oupDistinct false :groupingSets <> :havingQual <> :windowClau
se <> :distinctClause <> :sortClause <> :limitOffset <> :limi
tCount <> :limitOption 0 :rowMarks <> :setOperations <> :cons
traintDeps <> :withCheckOptions <> :stmt_location -1 :stmt_le
n -1}

```

WARNING: ===== REWRITE INPUT =====

```

WARNING: {QUERY :commandType 1 :querySource 0 :canSetTag tru
e :utilityStmt <> :resultRelation 0 :hasAggs true :hasWindowF
uncs false :hasTargetSRFs false :hasSubLinks true :hasDistinc
tOn false :hasRecursive false :hasModifyingCTE false :hasForU
pdate false :hasRowSecurity false :hasGroupRTE true :isReturn
false :cteList <> :rtable ({RANGETBENTRY :alias <> :eref {AL
IAS :aliasname orders :colnames ("o_orderkey" "o_custkey" "o_
orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority"
"o_clerk" "o_shippriority" "o_comment")}} :rtekind 0 :relid 16
561 :inh true :relkind r :rellockmode 1 :perminfoindex 1 :tab
lesample <> :lateral false :inFromCl true :securityQuals <> }
{RANGETBENTRY :alias <> :eref {ALIAS :aliasname *GROUP* :col
names ("o_orderpriority")}} :rtekind 9 :groupexprs ({VAR :varn
o 1 :varattno 6 :vartype 25 :vartypmod -1 :varcollid 100 :var
nullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn
1 :varattnosyn 6 :location -1}) :lateral false :inFromCl fals
e :securityQuals <>}) :rteperminfos ({RTEPERMISSIONINFO :reli
d 16561 :inh true :requiredPerms 2 :checkAsUser 0 :selectedCo
ls (b 8 12 13) :insertedCols (b) :updatedCols (b)}) :jointree
{FROMEXPR :fromlist ({RANGETBLREF :rtindex 1}) :quals {BOOLEX
PR :boolop and :args ({OEXPR :opno 1098 :opfuncid 1090 :opre
sulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args
({VAR :varno 1 :varattno 5 :vartype 1082 :vartypmod -1 :varco
llid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0

```

```

:varnosyn 1 :varattnosyn 5 :location -1} {CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]}) :location -1} {OPEXPR :opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 5 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1} {OPEXPR :opno 1076 :opfuncid 2071 :opresulttype 1114 :opretset false :opcollid 0 :inputcollid 0 :args ({CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]} {CONST :consttype 1186 :consttypmod 196607 :constcollid 0 :constlen 16 :constbyval false :constisnull false :location -1 :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]}) :location -1}) :location -1} {SUBLINK :subLinkType 0 :subLinkId 0 :testexpr <> :operName <> :subselect {QUERY :commandType 1 :querySource 0 :canSetTag true :utilityStmt <> :resultRelation 0 :hasAggs false :hasWindowFuncs false :hasTargetSRFs false :hasSubLinks false :hasDistinctOn false :hasRecursive false :hasModifyingCTE false :hasForUpdate false :hasRowSecurity false :hasGroupRTE false :isReturn false :cteList <> :rtable ({RANGETBLENTRY :alias <> :eref {ALIAS :aliasname lineitem :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")}} :rtekind 0 :relid 16566 :inh true :relkind r :rellockmode 1 :perminfoindex 1 :tablesample <> :lateral false :inFromCl true :securityQuals <>}) :rteperminfos ({RTEPERMISSIONINFO :relid 16566 :inh true :requiredPerms 2 :checkAsUser 0 :selectedCols (b 8 19 20) :insertedCols (b) :updatedCols (b)}) :jointree {FROMEXPR :fromlist ({RANGETBLREF :rtindex 1}) :quals {BOOLEXPR :boolop and :args ({OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrel

```

```

s (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattn
osyn 1 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 1 :
varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})
:location -1} {OPEXPR :opno 1095 :opfuncid 1087 :opresulttype
16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :va
rno 1 :varattno 12 :vartype 1082 :vartypmod -1 :varcollid 0 :
varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnos
yn 1 :varattnosyn 12 :location -1} {VAR :varno 1 :varattno 13
:vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b)
:varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1
3 :location -1}) :location -1}) :location -1}} :mergeActionLi
st <> :mergeTargetRelation 0 :mergeJoinCondition <> :targetLi
st ({TARGETENTRY :expr {CONST :consttype 23 :consttypmod -1 :
constcollid 0 :constlen 4 :constbyval true :constisnull false
:location -1 :constvalue 4 [ 1 0 0 0 0 0 0 0 ]} :resno 1 :res
name ?column? :ressortgroupref 0 :resorigtbl 0 :resorigcol 0
:resjunk false}) :override 0 :onConflict <> :returningOldAlia
s <> :returningNewAlias <> :returningList <> :groupClause <>
:groupDistinct false :groupingSets <> :havingQual <> :windowC
lause <> :distinctClause <> :sortClause <> :limitOffset <> :l
imitCount <> :limitOption 0 :rowMarks <> :setOperations <> :c
onstraintDeps <> :withCheckOptions <> :stmt_location -1 :stmt
_len -1} :location -1}) :location -1}} :mergeActionList <> :m
ergeTargetRelation 0 :mergeJoinCondition <> :targetList ({TAR
GETENTRY :expr {VAR :varno 2 :varattno 1 :vartype 25 :vartypm
od -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varr
eturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :resn
o 1 :resname o_orderpriority :ressortgroupref 1 :resorigtbl 1
6561 :resorigcol 6 :resjunk false} {TARGETENTRY :expr {AGGREF
:aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtr
anstype 0 :aggargtypes <> :aggdirectargs <> :args <> :aggorde
r <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic
false :aggkind n :aggpresorted false :agglevelsup 0 :aggsplit
0 :aggno -1 :aggtransno -1 :location -1} :resno 2 :resname or
der_count :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :res

```

```

junk false})) :override 0 :onConflict <> :returningOldAlias <>
:returningNewAlias <> :returningList <> :groupClause ({SORTGR
OUPCLAUSE :tleSortGroupRef 1 :eqop 98 :sortop 664 :reverse_so
rt false :nulls_first false :hashable true})) :groupDistinct f
alse :groupingSets <> :havingQual <> :windowClause <> :distin
ctClause <> :sortClause ({SORTGROUPCLAUSE :tleSortGroupRef 1
:eqop 98 :sortop 664 :reverse_sort false :nulls_first false :
hashable true})) :limitOffset <> :limitCount <> :limitOption 0
:rowMarks <> :setOperations <> :constraintDeps <> :withCheckO
ptions <> :stmt_location -1 :stmt_len -1}
WARNING: ===== RLS POLICY ADDED =====
WARNING: Table OID=16561
WARNING: ({SUBLINK :subLinkType 0 :subLinkId 0 :testexpr <>
:operName <> :subselect {QUERY :commandType 1 :querySource 0
:canSetTag true :utilityStmt <> :resultRelation 0 :hasAggs fa
lse :hasWindowFuncs false :hasTargetSRFs false :hasSubLinks f
alse :hasDistinctOn false :hasRecursive false :hasModifyingCT
E false :hasForUpdate false :hasRowSecurity false :hasGroupRT
E false :isReturn false :cteList <> :rtable ({RANGETBLENTRY :
alias {ALIAS :aliasname c :colnames <>} :eref {ALIAS :aliasna
me c :colnames ("c_custkey" "c_name" "c_address" "c_nationke
y" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")}} :rtekin
d 0 :relid 16546 :inh true :relkind r :rellockmode 1 :perminf
oindex 1 :tablesample <> :lateral false :inFromCl true :secur
ityQuals <>}} :rteperminfos ({RTEPERMISSIONINFO :relid 16546
:inh true :requiredPerms 2 :checkAsUser 0 :selectedCols (b 8
13) :insertedCols (b) :updatedCols (b))} :jointree {FROMEXPR
:fromlist ({RANGETBLREF :rtindex 1}) :quals {BOOLEXPRESS :boolop
and :args ({OPEXPRESS :opno 96 :opfuncid 65 :opresulttype 16 :op
retset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1
:varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullin
grels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :var
attnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 2
3 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup
1 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -
1}) :location -1} {OPEXPRESS :opno 674 :opfuncid 297 :opresultty

```

```

pe 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR
:varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0
:varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varno
syn 1 :varattnosyn 6 :location -1} {FUNCEXPR :funcid 316 :fun
cresulttype 701 :funcresultset false :funcvariadic false :funcfo
rmat 2 :funccollid 0 :inputcollid 0 :args ({CONST :consttype
23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval tru
e :constisnull false :location -1 :constvalue 4 [ -120 19 0 0
0 0 0 0 ]}) :location -1}) :location -1}) :location -1}) :mer
geActionList <> :mergeTargetRelation 0 :mergeJoinCondition <>
:targetList ({TARGETENTRY :expr {CONST :consttype 23 :constty
pmod -1 :constcollid 0 :constlen 4 :constbyval true :constisn
ull false :location -1 :constvalue 4 [ 1 0 0 0 0 0 0 0 ]} :re
sno 1 :resname ?column? :ressortgroupref 0 :resorigtbl 0 :res
origcol 0 :resjunk false}) :override 0 :onConflict <> :return
ingOldAlias <> :returningNewAlias <> :returningList <> :group
Clause <> :groupDistinct false :groupingSets <> :havingQual <
> :windowClause <> :distinctClause <> :sortClause <> :limitOf
fset <> :limitCount <> :limitOption 0 :rowMarks <> :setOperat
ions <> :constraintDeps <> :withCheckOptions <> :stmt_locatio
n -1 :stmt_len -1} :location -1})

```

WARNING: ===== REWRITE OUTPUT =====

```

WARNING: ({QUERY :commandType 1 :querySource 0 :canSetTag tr
ue :utilityStmt <> :resultRelation 0 :hasAggs true :hasWindow
Funcs false :hasTargetSRFs false :hasSubLinks true :hasDistin
ctOn false :hasRecursive false :hasModifyingCTE false :hasFor
Update false :hasRowSecurity true :hasGroupRTE true :isReturn
false :cteList <> :rtable ({RANGETBLEENTRY :alias <> :eref {AL
IAS :aliasname orders :colnames ("o_orderkey" "o_custkey" "o_
orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority"
"o_clerk" "o_shippriority" "o_comment")}} :rtekind 0 :relid 16
561 :inh true :relkind r :rellockmode 1 :perminfoindex 1 :tab
lesample <> :lateral false :inFromCl true :securityQuals ({SU
BLINK :subLinkType 0 :subLinkId 0 :testexpr <> :operName <> :
subselect {QUERY :commandType 1 :querySource 0 :canSetTag tru
e :utilityStmt <> :resultRelation 0 :hasAggs false :hasWindow

```

```

Funcs false :hasTargetSRFs false :hasSubLinks false :hasDistinctOn false :hasRecursive false :hasModifyingCTE false :hasForUpdate false :hasRowSecurity false :hasGroupRTE false :isReturn false :cteList <> :rtable ({RANGETBENTRY :alias {ALIAS :aliasname c :colnames <>} :eref {ALIAS :aliasname c :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")}} :rtekind 0 :relid 16546 :inh true :relkind r :rellockmode 1 :perminfoindex 1 :tablesample <> :lateral false :inFromCl true :securityQuals <>}) :rtperminfos ({RTEPERMISSIONINFO :relid 16546 :inh true :requiredPerms 2 :checkAsUser 0 :selectedCols (b 8 13) :insertedCols (b) :updatedCols (b)}) :jointree {FROMEXPR :fromlist ({RANGETBLREF :rtindex 1}) :quals {BOOLEXPRESS :boolop and :args ({OPEXPRESS :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevels up 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevels up 1 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1}) :location -1} {OPEXPRESS :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevels up 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {FUNCEXPRESS :funcid 316 :funcresulttype 701 :func retset false :funcvariadic false :funcformat 2 :func collid 0 :inputcollid 0 :args ({CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ -120 19 0 0 0 0 0 0 ]}) :location -1}) :location -1}) :location -1}} :mergeActionList <> :mergeTargetRelation 0 :mergeJoinCondition <> :targetList ({TARGETENTRY :expr {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 1 0 0 0 0 0 0 0 ]} :resno 1 :resname ?column? :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}) :override 0 :onConflict <> :returningOldAlias <> :retu

```

```

rningNewAlias <> :returningList <> :groupClause <> :groupDist
inct false :groupingSets <> :havingQual <> :windowClause <> :
distinctClause <> :sortClause <> :limitOffset <> :limitCount
<> :limitOption 0 :rowMarks <> :setOperations <> :constraintD
eps <> :withCheckOptions <> :stmt_location -1 :stmt_len -1} :
location -1}} {RANGETBENTRY :alias <> :eref {ALIAS :aliasna
me *GROUP* :colnames ("o_orderpriority")} :rtekind 9 :grouper
prs ({VAR :varno 1 :varattno 6 :vartype 25 :vartypmod -1 :var
collid 100 :varnullingrels (b) :varlevelsup 0 :varreturningty
pe 0 :varnosyn 1 :varattnosyn 6 :location -1}) :lateral false
:inFromCl false :securityQuals <>}) :rteperminfos ({RTEPERMIS
SIONINFO :relid 16561 :inh true :requiredPerms 2 :checkAsUser
0 :selectedCols (b 8 12 13) :insertedCols (b) :updatedCols
(b)}) :jointree {FROMEXPR :fromlist ({RANGETBLREF :rtindex
1}) :quals {BOOLEXPRESS :boolop and :args ({OPEXPRESS :opno 1098 :o
pfuncid 1090 :opresulttype 16 :opretset false :opcollid 0 :in
putcollid 0 :args ({VAR :varno 1 :varattno 5 :vartype 1082 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :
varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1}
{CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constl
en 4 :constbyval true :constisnull false :location -1 :constv
alue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]}) :location -1} {OPEXPRESS :
opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false :op
collid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 5 :var
type 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :var
levelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :lo
cation -1} {OPEXPRESS :opno 1076 :opfuncid 2071 :opresulttype 11
14 :opretset false :opcollid 0 :inputcollid 0 :args ({CONST :
consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :co
nstbyval true :constisnull false :location -1 :constvalue 4 [
113 -9 -1 -1 -1 -1 -1 -1 ]} {CONST :consttype 1186 :consttypm
od 196607 :constcollid 0 :constlen 16 :constbyval false :cons
tisnull false :location -1 :constvalue 16 [ 0 0 0 0 0 0 0 0 0
0 0 0 3 0 0 0 ]}) :location -1}) :location -1} {SUBLINK :subl
inkType 0 :subLinkId 0 :testexpr <> :operName <> :subselect
{QUERY :commandType 1 :querySource 0 :canSetTag true :utility

```

```

Stmt <> :resultRelation 0 :hasAggs false :hasWindowFuncs false
:hasTargetSRFs false :hasSubLinks false :hasDistinctOn false
:hasRecursive false :hasModifyingCTE false :hasForUpdate false
:hasRowSecurity false :hasGroupRTE false :isReturn false
:cteList <> :rtable ({RANGETBLENTRY :alias <> :eref {ALIAS :a
aliasname lineitem :colnames ("l_orderkey" "l_partkey" "l_supp
key" "l_linenummer" "l_quantity" "l_extendedprice" "l_discoun
t" "l_tax" "l_returnflag" "l_linestatus" "l_shipdate" "l_comm
itdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comm
ent")}} :rtekind 0 :relid 16566 :inh true :relkind r :rellockm
ode 1 :perminfoindex 1 :tablesample <> :lateral false :inFrom
Cl true :securityQuals <>)) :rteperminfos ({RTEPERMISSIONINFO
:relid 16566 :inh true :requiredPerms 2 :checkAsUser 0 :selec
tedCols (b 8 19 20) :insertedCols (b) :updatedCols (b)}) :joi
ntree {FROMEXPR :fromlist ({RANGETBLREF :rtindex 1}) :quals
{BOOLEXPRESS :boolop and :args ({OPEXPRESS :opno 96 :opfuncid 65 :o
presulttype 16 :opretset false :opcollid 0 :inputcollid 0 :ar
gs ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varc
ollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype
0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :var
attno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrel
s (b) :varlevelsup 1 :varreturningtype 0 :varnosyn 1 :varattn
osyn 1 :location -1}) :location -1} {OPEXPRESS :opno 1095 :opfun
cid 1087 :opresulttype 16 :opretset false :opcollid 0 :inputc
ollid 0 :args ({VAR :varno 1 :varattno 12 :vartype 1082 :vart
ypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :var
returningtype 0 :varnosyn 1 :varattnosyn 12 :location -1} {VA
R :varno 1 :varattno 13 :vartype 1082 :vartypmod -1 :varcolli
d 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :v
arnosyn 1 :varattnosyn 13 :location -1}) :location -1}) :loca
tion -1}} :mergeActionList <> :mergeTargetRelation 0 :mergeJo
inCondition <> :targetList ({TARGETENTRY :expr {CONST :constt
ype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval
true :constisnull false :location -1 :constvalue 4 [ 1 0 0 0
0 0 0 0 ]} :resno 1 :resname ?column? :ressortgroupref 0 :res
origtbl 0 :resorigcol 0 :resjunk false}) :override 0 :onConfl

```

```

ict <> :returningOldAlias <> :returningNewAlias <> :returning
List <> :groupClause <> :groupDistinct false :groupingSets <>
:havingQual <> :windowClause <> :distinctClause <> :sortClau
se <> :limitOffset <> :limitCount <> :limitOption 0 :rowMarks
<> :setOperations <> :constraintDeps <> :withCheckOptions <>
:stmt_location -1 :stmt_len -1} :location -1}) :location -1}}
:mergeActionList <> :mergeTargetRelation 0 :mergeJoinConditio
n <> :targetList ({TARGETENTRY :expr {VAR :varno 2 :varattno
1 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels
(b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnos
yn 1 :location -1} :resno 1 :resname o_orderpriority :ressort
grouppref 1 :resorigtbl 16561 :resorigcol 6 :resjunk false} {T
ARGETENTRY :expr {AGGREF :aggfnoid 2803 :aggtype 20 :aggcolli
d 0 :inputcollid 0 :aggtranstype 0 :aggargtypes <> :aggdirect
args <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :
aggstar true :aggvariadic false :aggkind n :aggpresorted fals
e :agglevelsup 0 :aggsplit 0 :aggno -1 :aggtransno -1 :locati
on -1} :resno 2 :resname order_count :ressortgrouppref 0 :reso
rigtbl 0 :resorigcol 0 :resjunk false}) :override 0 :onConfli
ct <> :returningOldAlias <> :returningNewAlias <> :returningL
ist <> :groupClause ({SORTGROUPCLAUSE :tleSortGroupRef 1 :eqo
p 98 :sortop 664 :reverse_sort false :nulls_first false :hash
able true}) :groupDistinct false :groupingSets <> :havingQual
<> :windowClause <> :distinctClause <> :sortClause ({SORTGROU
PCLAUSE :tleSortGroupRef 1 :eqop 98 :sortop 664 :reverse_sort
false :nulls_first false :hashable true}) :limitOffset <> :li
mitCount <> :limitOption 0 :rowMarks <> :setOperations <> :co
nstraintDeps <> :withCheckOptions <> :stmt_location -1 :stmt_
len -1})

```

WARNING: ===== PLANNER INPUT =====

```

WARNING: {QUERY :commandType 1 :querySource 0 :canSetTag tru
e :utilityStmt <> :resultRelation 0 :hasAggs true :hasWindowF
uncs false :hasTargetSRFs false :hasSubLinks true :hasDistinc
tOn false :hasRecursive false :hasModifyingCTE false :hasForU
pdate false :hasRowSecurity true :hasGroupRTE true :isReturn
false :cteList <> :rtable ({RANGETBLEENTRY :alias <> :eref {AL

```

```

IAS :aliasname orders :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")) :rtekind 0 :relid 16561 :inh true :relkind r :rellockmode 1 :perminfoindex 1 :tablesample <> :lateral false :inFromCl true :securityQuals ({SUBLINK :subLinkType 0 :subLinkId 0 :testexpr <> :operName <> :subselect {QUERY :commandType 1 :querySource 0 :canSetTag true :utilityStmt <> :resultRelation 0 :hasAggs false :hasWindowFuncs false :hasTargetSRFs false :hasSubLinks false :hasDistinctOn false :hasRecursive false :hasModifyingCTE false :hasForUpdate false :hasRowSecurity false :hasGroupRTE false :isReturn false :cteList <> :rtable ({RANGETBENTRY :alias {ALIAS :aliasname c :colnames <>} :eref {ALIAS :aliasname c :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment"))} :rtekind 0 :relid 16546 :inh true :relkind r :rellockmode 1 :perminfoindex 1 :tablesample <> :lateral false :inFromCl true :securityQuals <>}) :rtperminfos ({RTEPERMISSIONINFO :relid 16546 :inh true :requiredPerms 2 :checkAsUser 0 :selectedCols (b 8 13) :insertedCols (b) :updatedCols (b)}) :jointree {FROMEXPR :fromlist ({RANGETBLREF :rtindex 1}) :quals {BOOLEXPRESS :boolop and :args ({OPEXPRESS :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 1 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :location -1}) :location -1} {OPEXPRESS :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {FUNCEXPRESS :funcid 316 :funcresulttype 701 :funcresultset false :funcvariadic false :funcformat 2 :funccollid 0 :inputcollid 0 :args ({CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false

```

```

:location -1 :constvalue 4 [ -120 19 0 0 0 0 0 0 ]}) :location
n -1}) :location -1}) :location -1}) :mergeActionList <> :mer
geTargetRelation 0 :mergeJoinCondition <> :targetList ({TARGE
TENTRY :expr {CONST :consttype 23 :consttypmod -1 :constcolli
d 0 :constlen 4 :constbyval true :constisnull false :location
-1 :constvalue 4 [ 1 0 0 0 0 0 0 0 ]} :resno 1 :resname ?colu
mn? :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk f
alse}) :override 0 :onConflict <> :returningOldAlias <> :retu
rningNewAlias <> :returningList <> :groupClause <> :groupDist
inct false :groupingSets <> :havingQual <> :windowClause <> :
distinctClause <> :sortClause <> :limitOffset <> :limitCount
<> :limitOption 0 :rowMarks <> :setOperations <> :constraintD
eps <> :withCheckOptions <> :stmt_location -1 :stmt_len -1} :
location -1})} {RANGETBENTRY :alias <> :eref {ALIAS :aliasna
me *GROUP* :colnames ("o_orderpriority")} :rtekind 9 :groupe
xprs ({VAR :varno 1 :varattno 6 :vartype 25 :vartypmod -1 :var
collid 100 :varnullingrels (b) :varlevelsup 0 :varreturningty
pe 0 :varnosyn 1 :varattnosyn 6 :location -1}) :lateral false
:inFromCl false :securityQuals <>}) :rteperminfos ({RTEPERMIS
SIONINFO :relid 16561 :inh true :requiredPerms 2 :checkAsUser
0 :selectedCols (b 8 12 13) :insertedCols (b) :updatedCols
(b)}) :jointree {FROMEXPR :fromlist ({RANGETBLREF :rtindex
1}) :quals {BOOLEXPRESS :boolop and :args ({OPEXPRESS :opno 1098 :o
pfuncid 1090 :opresulttype 16 :opretset false :opcollid 0 :in
putcollid 0 :args ({VAR :varno 1 :varattno 5 :vartype 1082 :v
artypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :
varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1}
{CONST :consttype 1082 :consttypmod -1 :constcollid 0 :constl
en 4 :constbyval true :constisnull false :location -1 :constv
alue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]}) :location -1} {OPEXPRESS :
opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false :op
collid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 5 :var
type 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :var
levelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :lo
cation -1} {OPEXPRESS :opno 1076 :opfuncid 2071 :opresulttype 11
14 :opretset false :opcollid 0 :inputcollid 0 :args ({CONST :

```

```

consttype 1082 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [
113 -9 -1 -1 -1 -1 -1 -1 ]} {CONST :consttype 1186 :consttypmod 196607 :constcollid 0 :constlen 16 :constbyval false :constisnull false :location -1 :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]}) :location -1}) :location -1} {SUBLINK :sublinkType 0 :subLinkId 0 :testexpr <> :operName <> :subselect {QUERY :commandType 1 :querySource 0 :canSetTag true :utilityStmt <> :resultRelation 0 :hasAggs false :hasWindowFuncs false :hasTargetSRFs false :hasSubLinks false :hasDistinctOn false :hasRecursive false :hasModifyingCTE false :hasForUpdate false :hasRowSecurity false :hasGroupRTE false :isReturn false :cteList <> :rtable ({RANGETBENTRY :alias <> :eref {ALIAS :aliasname lineitem :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")}} :rtekind 0 :relid 16566 :inh true :relkind r :rellockmode 1 :perminfoindex 1 :tablesample <> :lateral false :inFromCl true :securityQuals <>}} :rteperminfos ({RTEPERMISSIONINFO :relid 16566 :inh true :requiredPerms 2 :checkAsUser 0 :selectedCols (b 8 19 20) :insertedCols (b) :updatedCols (b)}) :join tree {FROMEXPR :fromlist ({RANGETBLREF :rtindex 1}) :quals {BOOLEXPRESS :boolop and :args ({OPEXPRESS :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 1 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1}) :location -1} {OPEXPRESS :opno 1095 :opfuncid 1087 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 12 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 12 :location -1} {VAR :varno 1 :varattno 13 :vartype 1082 :vartypmod -1 :varcollid

```

```

d 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 13 :location -1}} :location -1}} :location -1}} :mergeActionList <> :mergeTargetRelation 0 :mergeJoinCondition <> :targetList ({TARGETENTRY :expr {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [ 1 0 0 0 0 0 0 0 ]} :resno 1 :resname ?column? :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}} :override 0 :onConflict <> :returningOldAlias <> :returningNewAlias <> :returningList <> :groupClause <> :groupDistinct false :groupingSets <> :havingQual <> :windowClause <> :distinctClause <> :sortClause <> :limitOffset <> :limitCount <> :limitOption 0 :rowMarks <> :setOperations <> :constraintDeps <> :withCheckOptions <> :stmt_location -1 :stmt_len -1} :location -1}} :location -1}} :mergeActionList <> :mergeTargetRelation 0 :mergeJoinCondition <> :targetList ({TARGETENTRY :expr {VAR :varno 2 :varattno 1 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :resno 1 :resname o_orderpriority :ressortgroupref 1 :resorigtbl 16561 :resorigcol 6 :resjunk false} {TARGETENTRY :expr {AGGREF :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 0 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <> :aggstar true :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno -1 :aggtransno -1 :location -1} :resno 2 :resname order_count :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}} :override 0 :onConflict <> :returningOldAlias <> :returningNewAlias <> :returningList <> :groupClause ({SORTGROUPCLAUSE :tleSortGroupRef 1 :eqop 98 :sortop 664 :reverse_sort false :nulls_first false :hashable true}} :groupDistinct false :groupingSets <> :havingQual <> :windowClause <> :distinctClause <> :sortClause ({SORTGROUPCLAUSE :tleSortGroupRef 1 :eqop 98 :sortop 664 :reverse_sort false :nulls_first false :hashable true}} :limitOffset <> :limitCount <> :limitOption 0 :rowMarks <> :setOperations <> :constraintDeps <> :withCheckOptions <> :stmt_location -1 :stmt_

```

```

len -1}
WARNING:  ===== FINAL PLAN =====
WARNING:  {AGG :plan.disabled_nodes 0 :plan.startup_cost 8883
274415.708885 :plan.total_cost 8883274719.763882 :plan.plan_r
ows 5 :plan.plan_width 17 :plan.parallel_aware false :plan.pa
rallel_safe false :plan.async_capable false :plan.plan_node_i
d 0 :plan.targetlist ({TARGETENTRY :expr {VAR :varno -2 :vara
ttno 1 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingre
ls (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 0 :varatt
nosyn 0 :location -1} :resno 1 :resname o_orderpriority :ress
ortgroupref 1 :resorigtbl 16561 :resorigcol 6 :resjunk false}
{TARGETENTRY :expr {AGGREG :aggfnoid 2803 :aggtype 20 :aggcol
lid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdir
ectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilter <
> :aggstar true :aggvariadic false :aggkind n :aggpresorted f
alse :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :locat
ion -1} :resno 2 :resname order_count :ressortgroupref 0 :res
origtbl 0 :resorigcol 0 :resjunk false}) :plan.qual <> :plan.
lefttree {SORT :plan.disabled_nodes 0 :plan.startup_cost 8883
274415.708885 :plan.total_cost 8883274517.043884 :plan.plan_r
ows 40534 :plan.plan_width 9 :plan.parallel_aware false :pla
n.parallel_safe false :plan.async_capable false :plan.plan_no
de_id 1 :plan.targetlist ({TARGETENTRY :expr {VAR :varno -2 :
varattno 1 :vartype 25 :vartypmod -1 :varcollid 100 :varnulli
ngrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :va
rattnosyn 6 :location -1} :resno 1 :resname <> :ressortgroupr
ef 1 :resorigtbl 0 :resorigcol 0 :resjunk false}) :plan.qual
<> :plan.lefttree {HASHJOIN :join.plan.disabled_nodes 0 :joi
n.plan.startup_cost 212549.3375 :join.plan.total_cost 8883271
313.470623 :join.plan.plan_rows 40534 :join.plan.plan_width 9
:join.plan.parallel_aware false :join.plan.parallel_safe fals
e :join.plan.async_capable false :join.plan.plan_node_id 2 :j
oin.plan.targetlist ({TARGETENTRY :expr {VAR :varno -2 :varat
tno 1 :vartype 25 :vartypmod -1 :varcollid 100 :varnullingrel
s (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattn
osyn 6 :location -1} :resno 1 :resname <> :ressortgroupref 1

```

```

:resorigtbl 0 :resorigcol 0 :resjunk false})) :join.plan.qual
<> :join.plan.lefttree {SEQSCAN :scan.plan.disabled_nodes 0 :
scan.plan.startup_cost 0 :scan.plan.total_cost 8883048508 :sc
an.plan.plan_rows 144199 :scan.plan.plan_width 13 :scan.plan.
parallel_aware false :scan.plan.parallel_safe false :scan.pla
n.async_capable false :scan.plan.plan_node_id 3 :scan.plan.ta
rgetlist ({TARGETENTRY :expr {VAR :varno 1 :varattno 6 :varty
pe 25 :vartypmod -1 :varcollid 100 :varnullingrels (b) :varle
velsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :loca
tion -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl
0 :resorigcol 0 :resjunk false} {TARGETENTRY :expr {VAR :varn
o 1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnu
llingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1
:varattnosyn 1 :location -1} :resno 2 :resname <> :ressortgro
upref 0 :resorigtbl 0 :resorigcol 0 :resjunk false})) :scan.pl
an.qual ({OPEXPR :opno 1098 :opfuncid 1090 :opresulttype 16 :
opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno
1 :varattno 5 :vartype 1082 :vartypmod -1 :varcollid 0 :varnu
llingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1
:varattnosyn 5 :location -1} {CONST :consttype 1082 :consttyp
mod -1 :constcollid 0 :constlen 4 :constbyval true :constisnu
ll false :location -1 :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -
1 ]}) :location -1} {SUBPLAN :subLinkType 2 :testexpr {OPEXPR
:opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcol
lid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 2 :varty
pe 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevel
sup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 2 :locatio
n -1} {PARAM :paramkind 1 :paramid 1 :paramtype 23 :paramtypm
od -1 :paramcollid 0 :location -1})) :location -1} :paramIds
(i 1) :plan_id 2 :plan_name SubPlan\ 2 :firstColType 23 :firs
tColTypmod -1 :firstColCollation 0 :useHashTable true :unknow
nEqFalse true :parallel_safe true :setParam <> :parParam <> :
args <> :startup_cost 5716.52 :per_call_cost 0.0025} {OPEXPR
:opno 2345 :opfuncid 2338 :opresulttype 16 :opretset false :o
pcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 5 :va
rtype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :va

```

```

rlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 5 :location -1} {CONST :consttype 1114 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]}) :location -1}) :scan.plan.lefttree <> :scan.plan.righttree <> :scan.plan.initPlan <> :scan.plan.extParam (b) :scan.plan.allParam (b) :scan.scanreldid 1} :join.plan.righttree {HASH :plan.disabled_nodes 0 :plan.startup_cost 179731.5 :plan.total_cost 179731.5 :plan.plan_rows 2000307 :plan.plan_width 4 :plan.parallel_aware false :plan.parallel_safe true :plan.async_capable false :plan.plan_node_id 4 :plan.targetlist ({TARGETENTRY :expr {VAR :varno -2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1} :resno 1 :resname <> :resortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}) :plan.qual <> :plan.lefttree {SEQSCAN :scan.plan.disabled_nodes 0 :scan.plan.startup_cost 0 :scan.plan.total_cost 179731.5 :scan.plan.plan_rows 2000307 :scan.plan.plan_width 4 :scan.plan.parallel_aware false :scan.plan.parallel_safe true :scan.plan.async_capable false :scan.plan.plan_node_id 5 :scan.plan.targetlist ({TARGETENTRY :expr {VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1} :resno 1 :resname <> :resortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}) :scan.plan.qual ({OPEXPR :opno 1095 :opfuncid 1087 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 3 :varattno 12 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 12 :location -1} {VAR :varno 3 :varattno 13 :vartype 1082 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 13 :location -1}) :location -1}) :scan.plan.lefttree <> :scan.plan.righttree <> :scan.plan.initPlan <> :scan.plan.extParam (b) :scan.plan.allParam (b) :scan.scanreldid 3} :plan.righttree <> :plan.initPlan <> :plan.extParam (b) :plan.allParam (b) :hashkeys ({VAR :varno -2 :

```

```

varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1}) :skewTable 16561 :skewColumn 1 :skewInherit false :rows_total 0} :join.plan.initPlan <> :join.plan.extParam (b) :join.plan.allParam (b) :join.jointype 4 :join.inner_unique false :join.joinqual <> :hashclauses ({OPEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno -2 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1} {VAR :varno -1 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1}) :location -1}) :hashoperators (o 96) :hashcollations (o 0) :hashkeys ({VAR :varno -2 :varattno 2 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 1 :location -1})} :plan.righttree <> :plan.initPlan <> :plan.extParam (b) :plan.allParam (b) :numCols 1 :sortColIdx ( 1) :sortOperators ( 664) :collations ( 100) :nullsFirst ( false)} :plan.righttree <> :plan.initPlan <> :plan.extParam (b) :plan.allParam (b) :aggstrategy 1 :aggssplit 0 :numCols 1 :grpColIdx ( 1) :grpOperators ( 98) :grpCollations ( 100) :numGroups 5 :transitionSpace 0 :aggParams (b) :groupingSets <> :chain <>}

```

QUERY PLAN

```

-----
-----
-----
-----
GroupAggregate  (cost=8883274415.71..8883274719.76 rows=5 width=17) (actual time=2710.176..2712.097 rows=5.00 loops=1)
  Output: orders.o_orderpriority, count(*)
  Group Key: orders.o_orderpriority
  Buffers: shared hit=3 read=134400, temp read=14748 written=14748

```

```

-> Sort (cost=8883274415.71..8883274517.04 rows=40534 width=9) (actual time=2709.691..2710.565 rows=23317.00 loops=1)
    Output: orders.o_orderpriority
    Sort Key: orders.o_orderpriority
    Sort Method: quicksort Memory: 769kB
    Buffers: shared hit=3 read=134400, temp read=14748 written=14748
-> Hash Semi Join (cost=212549.34..8883271313.47 rows=40534 width=9) (actual time=1946.181..2702.654 rows=23317.00 loops=1)
    Output: orders.o_orderpriority
    Hash Cond: (orders.o_orderkey = lineitem.l_orderkey)
    Buffers: shared read=134400, temp read=14748 written=14748
-> Seq Scan on public.orders (cost=0.00..8883048508.00 rows=144199 width=13) (actual time=47.493..343.775 rows=25383.00 loops=1)
    Output: orders.o_orderpriority, orders.o_orderkey
    Filter: ((orders.o_orderdate >= '1994-01-01'::date) AND (ANY (orders.o_custkey = (hashed SubPlan 2).coll)) AND (orders.o_orderdate < '1994-04-01 00:00:00'::timestamp without time zone))
    Rows Removed by Filter: 1474617
    Buffers: shared read=29680
    SubPlan 2
-> Seq Scan on public.customer c (cost=0.00..5547.00 rows=67808 width=4) (actual time=0.168..32.956 rows=67989.00 loops=1)
    Output: c.c_custkey
    Filter: (c.c_acctbal > '5000'::double precision)
    Rows Removed by Filter: 82011
    Buffers: shared read=3672
-> Hash (cost=179731.50..179731.50 rows=2000

```

```

307 width=4) (actual time=1894.585..1894.585 rows=3793296.00
loops=1)
           Output: lineitem.l_orderkey
           Buckets: 262144 (originally 262144)  Bat
ches: 32 (originally 16)  Memory Usage: 6223kB
           Buffers: shared read=104720, temp writte
n=10750
           -> Seq Scan on public.lineitem (cost=
0.00..179731.50 rows=2000307 width=4) (actual time=0.461..119
1.650 rows=3793296.00 loops=1)
           Output: lineitem.l_orderkey
           Filter: (lineitem.l_commitdate < l
ineitem.l_receiptdate)
           Rows Removed by Filter: 2207919
           Buffers: shared read=104720

Planning:
  Buffers: shared hit=187 read=32
Planning Time: 10.932 ms
Execution Time: 2714.287 ms
(37 rows)

tpch=>

```

```

# 1. Compile the code (utilizes all available CPU cores)
make -j$(sysctl -n hw.ncpu)

# 2. Install the compiled binaries to your custom location (/
Users/adityadave/pgsql)
make install

# 3. Restart the PostgreSQL server (uses your data directory
/Users/adityadave/pgdata)
/Users/adityadave/pgsql/bin/pg_ctl -D /Users/adityadave/pgdat
a restart

```

